# Jackson Kaunismaa

Toronto, Canada  jackson-kaunismaa  JacksonKaunismaa

## Education

**University of Toronto** *Sept 2019 – June 2024*
*BASc in Engineering Science, Machine Intelligence*

- Major GPA: 4.0/4.0
- Cumulative GPA: 3.7/4.0
- **Coursework:** Artificial Intelligence; Machine Intelligence, Software and Neural Networks; Probabilistic Reasoning; Decision Support Systems; Matrix Algebra; Nonlinear Optimization; Mathematical Programming

## Experience

**Research Fellow** *Berkeley / London / Toronto*
*ML Alignment & Theory Scholars* *Jan 2025 – present*

- Developed method to elicit harmful capabilities in open-source models by fine-tuning on frontier model outputs from adjacent, non-refused domains. Recovered ∼40% of the capability gap on hazardous chemical synthesis tasks.
- Investigated whether prompt injection and jailbreaking documents improve frontier model performance on monitor-bypassing sabotage tasks. Found no significant effect, suggesting these documents can be filtered from pretraining data without capability loss.
- Developing realistic scheming evaluations and investigating effects of character training on scheming propensities and auditability. *In progress.*

**Research Volunteer** *Toronto, ON*
*Supervised Program for Alignment Research* *June 2024 – Feb 2025*

- Extended circuit discovery methods from Marks et al. to automatically identify sparse, interpretable circuits in GPT-2 using SAE features.
- Built infrastructure for testing scalable oversight protocols based on Kenton et al., including agentic tool-use pipelines to create reasoning gaps.

**ML Researcher** *Toronto, ON*
*Gene2Lead / University of Toronto* *Oct 2023 – Jan 2024*

- Built feature extraction pipeline from RCSB Protein Data Bank; trained XGBoost and CNN models to predict covalent reactivity of amino acid active sites, achieving 93% AUC

**Undergraduate Researcher** *Toronto, ON*
*Vector Institute (Supervised by Sanja Fidler)* *May 2023 – Aug 2023*

- Implemented Transformers from scratch with multi-GPU data parallelism; benchmarked position embedding schemes across compute scales

**ML Researcher** *Toronto, ON*
*University of Toronto (Supervised by Michael Guerzhoy)* *Nov 2022 – Aug 2023*

- Developed novel saliency method for CNNs on intensity-sensitive classification tasks; published at ICLR 2023

**Compiler Developer Intern** *Markham, ON*
*IBM Canada* *May 2022 – May 2023*

- Solved bugs and integrated changes to the CPython interpreter for IBM z/OS; integrated BLAS/LAPACK into z/OS builds of NumPy and SciPy

**ML Research Intern** *Toronto, ON*
*RBC Borealis AI (Supervised by Marcus Brubaker)* *July 2019 – Aug 2019*

- Developed novel convolution method for generative normalizing flows, extending emerging convolutions for texture synthesis

## Publications

**Eliciting Harmful Capabilities by Fine-Tuning on Safeguarded Outputs** *2026*

J. Kaunismaa, J. Hughes, C. Q. Knight, A. Griffin, M. Sharma, E. Jones

ICLR 2026 arXiv:2601.13528

**A Benchmark for Scalable Oversight Mechanisms** *2025*

A. Pallavi Sudhir, J. Kaunismaa, A. Panickssery

ICLR 2025 Bi-Align Workshop arXiv:2504.03731

**An Investigation into Energy Minimization Properties of MLP Features in LLMs** *May 2024*

J. Kaunismaa, V. Papyan

Undergraduate Thesis [PDF]

**How do ConvNets Understand Image Intensity?** *June 2023*

J. Kaunismaa, M. Guerzhoy

Tiny Papers @ ICLR 2023 arXiv:2306.00360

## Projects

**Sparse Circuit Discovery** *Code* *Writeup*

- Automatic circuit discovery in GPT-2 using SAE features; extended methods from Marks et al. to find sparse, interpretable circuits

**CUDA Evolution Simulator** *Code*

- High-performance evolution engine using custom CUDA kernels, OpenGL visualization, and PyTorch

**Transformer Benchmarking** *Code*

- Transformer implementation from scratch with multi-GPU data parallelism; comparison of position embeddings

**AlphaZero for Connect Four** *Code*

- Implementation of AlphaZero for 8x8 Connect Four with comparison to alpha-beta search

## Skills

**Languages & Frameworks:** Python, PyTorch, Inspect, C/C++, CUDA, Bash, Git, LaTeX

**ML:** Transformers, mechanistic interpretability, SAEs, evaluations, RL, normalizing flows

## References

**Erik Jones** – Research Scientist, Anthropic

**Erik Jenner** – Research Scientist, Google DeepMind

**Michael Guerzhoy** – Assistant Professor, University of Toronto